Training and testing role transfer in transformers & language models

Project proposal • Joel Vázquez López

Language technology | October 2026

The role problem

Same event

Marco chased Polo

Polo was chased by Marco

Actor: Marco in both sentences

Reversed event

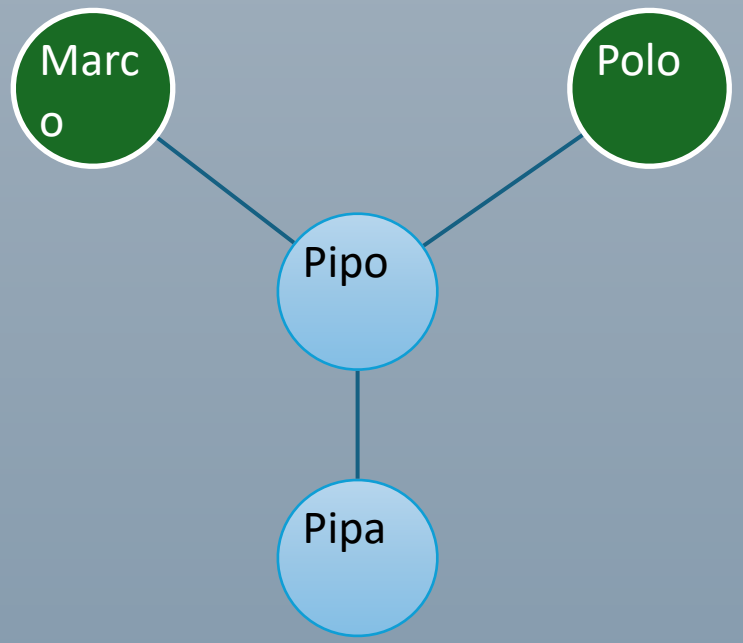
Marco chased Polo

Polo chased Marco.

Actor changes

Related work: Kim et al. (2020); Patil et al. (2024)

But here's the really interesting question:
What if the model has never seen Marco and Polo together before?



Both characters are familiar to the model, but it has never seen them together.

Can it still understand who is chasing whom?

The unseen-pair test



Not observed together during training

Test 1
Marco chased Polo
Actor: Marco

Test 2
Polo chased Marco
Actor: Polo

A

Research questions

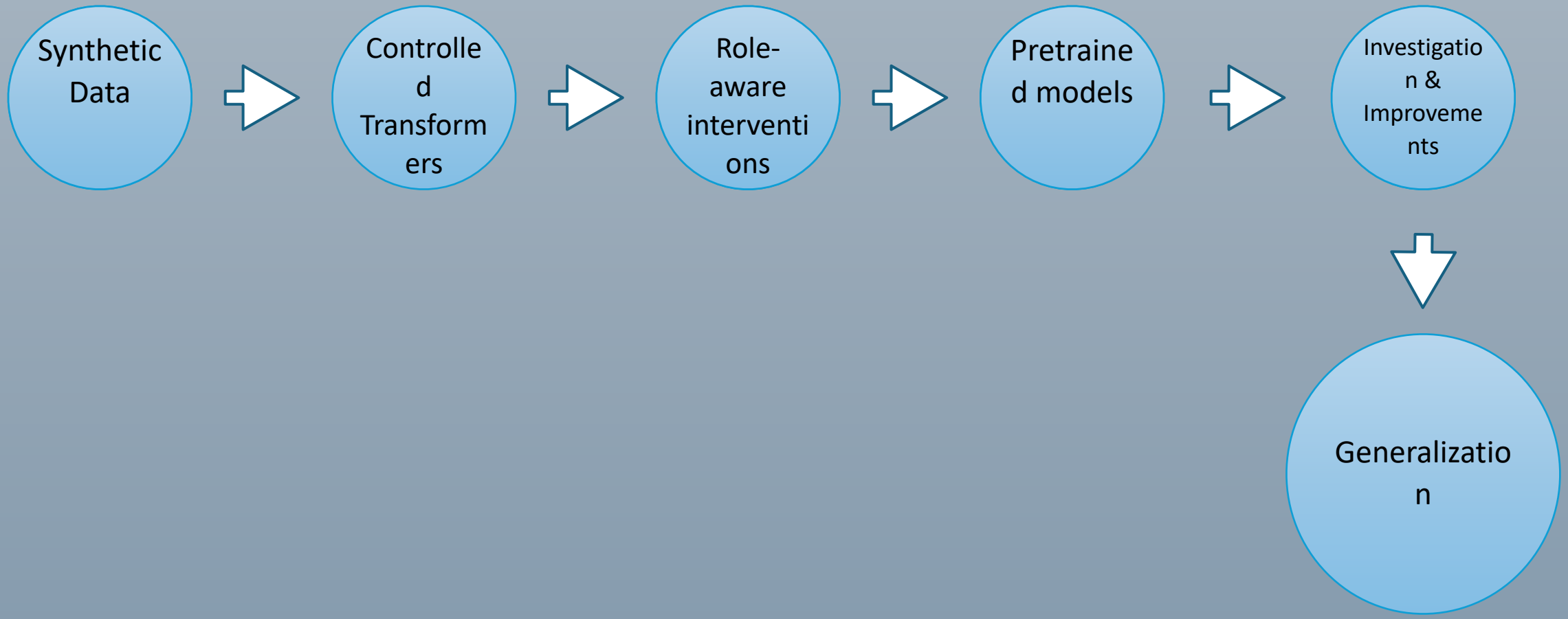
RQ1 When does a small Transformer transfer roles to a pair it never saw together?

RQ2 Can a role-aware training rule improve decisions on that missing pair?

RQ3 “Do pretrained language models identify semantic roles reliably across sentence forms and question formulations?”

RQ1 and RQ2 are the core study. RQ3 connects it to LLMs

From controlled experiments to language models



Real data · unseen prompts · more models

Controlled task: what changes?

Training

~10 characters
2 actions
~Roughly 20 observed pairs

Test

Unseen Pair
-Marco chased Polo
-Polo chased Marco

Choose the correct scene in both directions

What do we compare?

Different patterns of observed pairs
Ordinary training vs. role-aware training

From controlled tasks to real language

Sentence variation

Marco chased Polo.
Polo was chased by Marco.

Different structures, same actor: Marco

Question variation

Who performed the action?
Who did the chasing?

Different wording, same expected answer: Marco

Datasets

Controlled synthetic examples
Annotated real-language sentences

Multiple constructions, roles and question formulations

Models

Small controlled Transformers
Multiple pretrained language models

Compare robustness across model families and sizes

Evaluate accuracy and consistency on unseen prompts and sentences.

What do we hope to learn?

1. **Understand** —> Identify what enables or prevents reliable transfer of actor–patient roles.

2. **Improve** —> Investigate whether role-aware training and other targeted interventions increase robustness across linguistic variations.

3. **Validate** —> Test whether improvements generalize to unseen prompts, larger datasets, authentic sentences and different language models.